

Carbon Steel Grade 1045 Fact Sheet

AISI 1045 Steel, cold drawn, 19-32 mm (0.75-1.25 in) round

Subcategories: Carbon Steel; AISI 1000 Series Steel; Medium Carbon Steel

Close Analogs:

Key Words: AFNOR NF A35-553 XC45, AFNOR NF A35-554 XC48, DIN 1654 1.1192, DIN 1654 Cq45, DIN 17200 1.0503, AFNOR XC42, AFNOR XC42TS, AFNOR XC48TS, AFNOR NF A33-101 AF65C45, AFNOR NF A35-552 XC48H1, UNS G10450, ASTM A29, ASTM A108, ASTM A266 Class 3, ASTM A304, ASTM A311, ASTM A510, ASTM A519, AS 1442 K1045 (Australia), AS 1442 S1045, AS 1443 K1045, AS 1443 S1045, AS 1446 K1045, AS 1446 S1045, ASTM A568, ASTM A576, ASTM A682, ASTM A827, ASTM A830, FED QQ-S-635, FED QQ-S-700, FED QQ-W-461, MIL S-24093, MIL S-3039, BDS 6354 45G2A, BDS 6354 45G2K2, BDS 6354 45G2K3, GB 3078 45 (China), GB 3088 45, GB 699 45, YB 6 45B, DIN 17200 1.1191, DIN 17200 1.1201, DIN 17200 C45, DIN 17200 CK45, DIN 17200 Cm45, DIN 17200 GS-CK45, DIN 17212 1.1193, MIL S-46070, SAE J1397, SAE J403, SAE J412, BS 970 Part 1 060A47 (U.K), BS 970 Part 1 080A47, BS 970 Part 1 080M46, NBN 253-02 C45-3, NBN 253-06 C46, BDS 3492 45LI (Bulgaria), BDS 3492 45LII, BDS 3492 45LIII, BDS 5785 45, BDS 6354 45G2, ONORM M3108 C45SW (Austria), ONORM M3110 RC45, ONORM M3161 C45, NBN 253-02 C45-1 (Belgium), NBN 253-02 C45-2

Component	Weight %					
C	Max 0.50					
Fe	98.51 - 98.98					
Mn	0.60-0.90					
P	Max 0.040					
S	Max 0.050					
Physical Properties	Metric	English	Comments	Cold Head	Hot Forge	Wire Form
Density	7.85 g/cc	0.284 lb/in ³				
Mechanical Properties	Metric	English	Comments	Cold Head	Hot Forge	Wire Form
Hardness, Brinell	179	179				
Hardness, Knoop	200	200	Converted from Brinell hardness.			

Hardness, Rockwell B	88	88	Converted from Brinell hardness.			
Hardness, Vickers	188	188	Converted from Brinell hardness.			
Tensile Strength, Ultimate	625 MPa	90600 psi				
Tensile Strength, Yield	530 MPa	76900 psi				
Elongation at Break	12%	12%	In 50 mm			
Reduction of Area	35%	35%				
Modulus of Elasticity	206 GPa	29900 ksi				
Bulk Modulus	163 GPa	23600 ksi	Estimated from elastic modulus			
Poissons Ratio	0.29	0.29	Typical for steel			
Machinability	55.00%	55.00%	Based on AISI 1212 steel. as 100% machinability			
Shear Modulus	80.0 GPa	11600 ksi	Estimated from elastic			
Electrical Properties	Metric	English	Comments	Cold Head	Hot Forge	Wire Form
Electrical Resistivity	0.0000162 ohm-cm	0.0000162 ohm-cm				
	0.0000223 ohm-cm	0.0000223 ohm-cm				
Thermal Properties	Metric	English	Comments	Cold Head	Hot Forge	Wire Form
CTE, linear	11.5 µm/m-°C	6.39 µin/in-°F				
	Temperature 0.000 - 100 °C	Temperature 32.0 - 212 °F				
	13.0 µm/m-°C	7.22 µin/in-°F				
	Temperature 0.000 - 300 °C	Temperature 32.0 - 572 °F				
	14.0 µm/m-°C	7.78 µin/in-°F				
	Temperature 0.000 - 500 °C	Temperature 32.0 - 932 °F				
Specific Heat Capacity	0.486 J/g-°C	0.116 BTU/lb-°F	annealed			
	Temperature >=100 °C	Temperature >=212 °F				
	0.519 J/g-°C	0.124 BTU/lb-°F				

	Temperature 150 - 200 °C	Temperature 302 - 392 °F	
	0.586 J/g-°C	0.140 BTU/lb- °F	
	Temperature 350 - 400 °C	Temperature 662 - 752 °F	
Thermal Conductivity	49.8 W/m-K	346 BTU-in/hr- ft ² -°F	Typical steel